

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Drill	1	hr

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9702/33

October/November 2011

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You will be allowed to work with the apparatus for a maximum of one hour for each question.
You are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them.
You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be used only if it becomes necessary to do so. You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **10** printed pages and **2** blank pages.

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1 In this experiment you will determine the resistivity of a metal in the form of a wire.

- (a) (i) Measure and record the diameter d of the short sample of wire that is attached to the card. You may remove the wire from the card.

$d = \dots\dots\dots$ [1]

- (ii) Calculate the cross-sectional area A of the wire, in m^2 , using the formula

$$A = \frac{\pi d^2}{4}.$$

$A = \dots\dots\dots \text{m}^2$

- (b) (i) Using the wire that is taped to the metre rule, set up the circuit shown in Fig. 1.1. Close the switch.

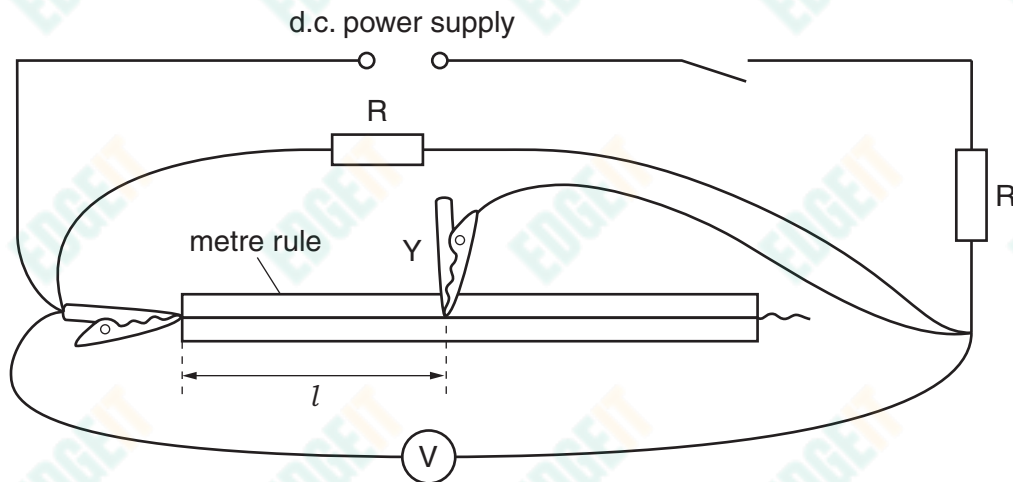


Fig. 1.1

- (ii) Position the crocodile clip Y approximately half-way along the wire.
(iii) Measure and record the length l of wire between the two crocodile clips and the voltmeter reading V .

$l = \dots\dots\dots$

$V = \dots\dots\dots$

[1]

- (c) By moving crocodile clip Y, change l and repeat (b)(iii) until you have six sets of readings of l and V for $l \geq 20$ cm. Include values of $\frac{1}{l}$ and $\frac{1}{V}$ in your table.

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[10]

- (d) (i) Plot a graph of $\frac{1}{V}$ on the y-axis against $\frac{1}{l}$ on the x-axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and y-intercept of this line.

gradient =

y-intercept =

[2]

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- (e) The quantities V and l are related by the equation

$$\frac{1}{V} = \frac{M}{l} + N$$

where M and N are constants.

- (i) Use your answers in (d)(iii) to determine values for M and N .

$$M = \dots\dots\dots \text{mV}^{-1}$$

$$N = \dots\dots\dots \text{V}^{-1}$$

[1]

- (ii) The resistivity ρ of the material of the wire, in Ωm , can be found using the relationship

$$\rho = \frac{5 \, A \, N}{M}.$$

Using your answers in (e)(i) and (a)(ii), calculate a value for ρ .

$$\rho = \dots\dots\dots \Omega\text{m} [1]$$

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Please turn over for Question 2.

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- 2 In this experiment you will investigate how the motion of a metre rule depends on the length of the string loops used to suspend it.

(a) Measure and record the width w of one of the metre rules, as shown in Fig. 2.1.

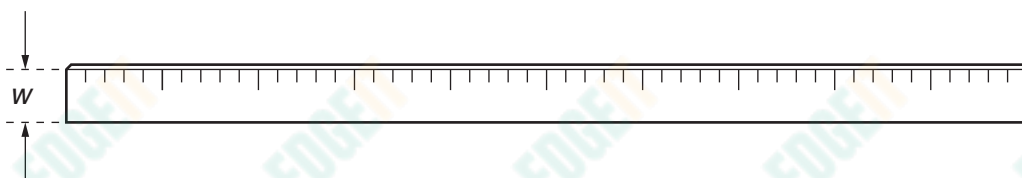


Fig. 2.1

$w =$ cm [1]

- (b) (i) Select the two longer pieces of string.
(ii) Tie the ends of one piece of string to make a loop.
(iii) Measure and record the length l of this loop, as shown in Fig. 2.2.

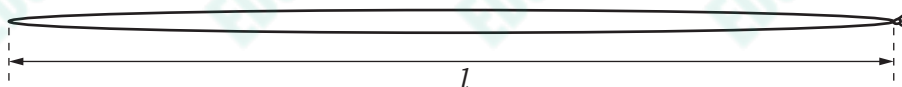


Fig. 2.2

$l =$ [1]

- (iv) Repeat (ii) with the other long piece of string. The length of this loop should be the same as that in (iii).

- (c) (i) Use the stands to set up the two metre rules and the two loops of string as shown in Fig. 2.3.

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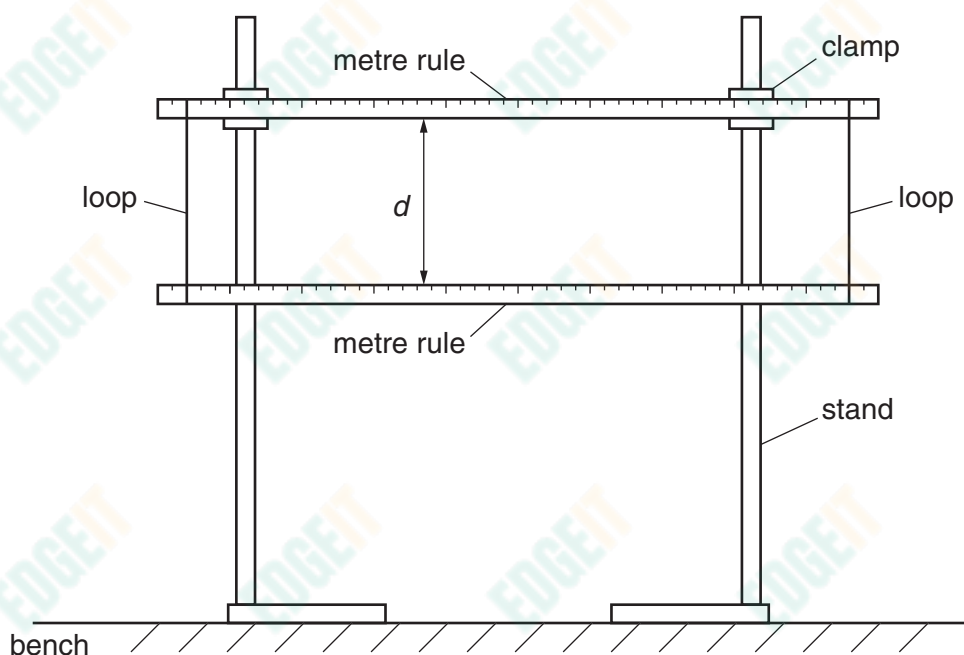


Fig. 2.3

The rules should be horizontal with the scale markings facing you.

The loops should be vertical, parallel to each other and placed at the 5 cm and 95 cm marks on both rules.

- (ii) Using your values in (a) and (b)(iii), determine the distance d using the relationship

$$d = l - 2w.$$

$d =$ [1]

- (iii) Estimate the percentage uncertainty in your value of d .

percentage uncertainty = [1]

- (d) Move the left end of the bottom rule towards you and the right end away from you. Release the rule and watch the movement. The left end of the rule will move away from you and back towards you, completing a swing. The time taken for one complete swing is T . By timing several of these complete swings, determine an accurate value for T .

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$$T = \dots\dots\dots \text{ s [2]}$$

- (e) Repeat (b), (c)(i), (c)(ii) and (d) for the shorter lengths of string.

$$l = \dots\dots\dots \text{ [1]}$$

$$d = \dots\dots\dots$$

$$T = \dots\dots\dots \text{ s [2]}$$

- (f) It is suggested that the relationship between T and d is

$$T^2 = kd$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$ [1]

- (ii) Justify the number of significant figures that you have given for your values of k .

.....

 [1]

- (iii) Explain whether your results support the suggested relationship.

.....

 [1]

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- (g) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

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[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.
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4.
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[4]

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